

A Proposed League of Legends Draft Tool

...

Roderic Moreno - DATA400

Presentation Outline:

- Project background and description
 - What is League?
 - Why is this project important?
 - Why did I choose this topic?
- Data
 - Data sources
 - Cleaning and processing

Project Background

- League of Legends is one of the largest online games
 - 130,000,000 active monthly players globally
- 5 major regional tier 1 leagues with 68 teams total
 - 16 across both Americas
 - 10 in Korea
 - 10 in Europe
 - 16 in China
 - 8 in Southeast Asia

Goals

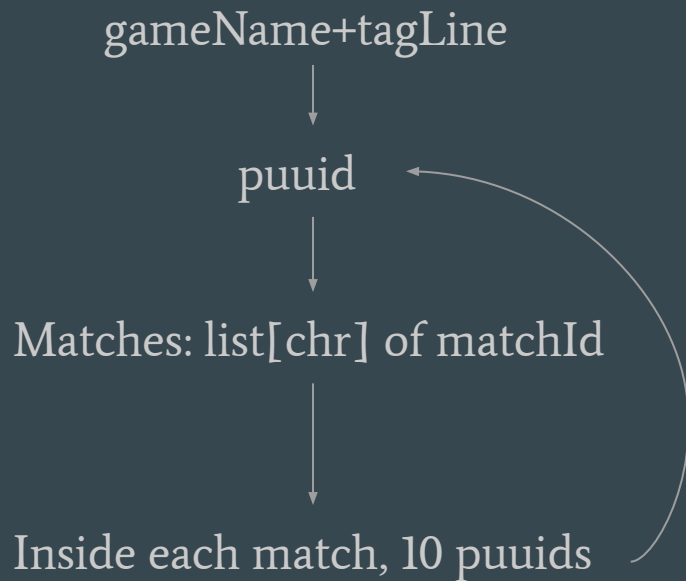
- Provide players with insight into what roles they have in certain situations
 - Allow testing for synergy with other players
- Predict the outcome of one of the early “breakpoints” inside each match

Method

- Collect match data from high ranks
- Create a theoretical model
 - Based on general players’ opinions, and pro-play statistics
- Fit a prediction model on game features
- Compare prediction with real drafts from professional games

Data Gathering

- Data pulled directly from the Riot Games API
 - `/riot/account/v1/accounts/by-riot-id/{gameName}/{tagLine}`
 - `/match/v5/matches/by-puuid`
 - `/match/v5/matches/{matchId}/timeline`
- Rate limit of 20 per second or 100 per 2 minutes
- ~7,600 matchIds pulled



Sample API Responses

RESPONSE BODY

```
{
  "puuid": "fqN5jmwEshO2AYHBMnHpbwLCmPC06VVpeZY5pvUN-SzqnmK4VyONc9JNp4fbuZ1vhjUIkmtkA7CKg",
  "gameName": "Jooba",
  "tagLine": "blues"
}
```

RESPONSE BODY

```
[
  "NA1_5473590148",
  "NA1_5473528810",
  "NA1_5473343060",
  "NA1_5472821088",
  "NA1_5472777192",
  "NA1_5472729929",
  "NA1_5471900773",
  "NA1_5471820649",
  "NA1_5470177749",
  "NA1_5468836372"
]
```

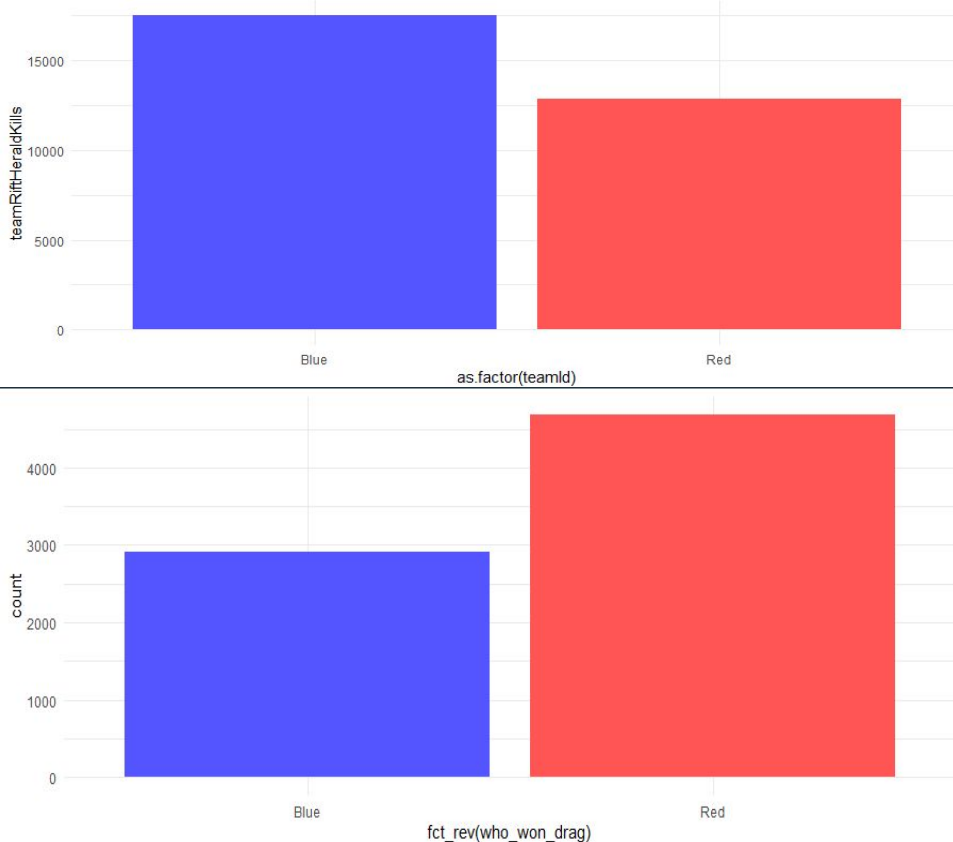
RESPONSE BODY

```
wardsguarded : 1
},
"champExperience": 19468,
"champLevel": 18,
"championId": 59,
"championName": "JarvanIV",
"championTransform": 0,
"commandPings": 8,
"consumablesPurchased": 5,
"damageDealtToBuildings": 1359,
"damageDealtToObjectives": 42180,
"damageDealtToTurrets": 1359,
"damageSelfMitigated": 57927,
"dangerPings": 0,
"deaths": 4,
"detectorWardsPlaced": 5,
"doubleKills": 1,
"dragonKills": 3,
"eligibleForProgression": true,
"enemyMissingPings": 23,
"enemyVisionPings": 0
```

Data Wrangling/EDA

- Timeline data taken from minutes 2-7
- Very imbalanced dataset
- Individual data to match data

```
processed_data      75960 obs. of 27 variables
$ matchId           : chr [1:75960] "NA1_5569684064" "NA1_5569684064" "NA1_5569684064" ...
$ participantId      : int [1:75960] 1 2 3 4 5 6 7 8 9 10 ...
$ championName       : chr [1:75960] "Akali" "Nunu" "Urgot" "Jinx" ...
$ teamId             : int [1:75960] 100 100 100 100 100 200 200 200 200 200 ...
$ teamPosition       : chr [1:75960] "TOP" "JUNGLE" "MIDDLE" "BOTTOM" ...
$ totalMinionsKilled : int [1:75960] 316 26 259 267 35 292 2 218 300 30 ...
$ kills              : int [1:75960] 10 7 7 5 2 1 4 7 12 5 ...
$ assists            : int [1:75960] 5 15 13 9 15 6 11 6 11 20 ...
$ win                : num [1:75960] 1 1 1 1 1 0 0 0 0 ...
$ firstTowerKill     : num [1:75960] 0 0 0 0 0 0 0 0 0 ...
$ firstBloodKill     : num [1:75960] 0 0 0 0 0 0 0 0 0 ...
$ firstBloodAssist   : num [1:75960] 0 0 0 0 0 0 0 0 0 ...
$ quickSoloKills     : int [1:75960] 1 0 0 0 0 0 0 0 0 ...
$ quickFirstTurret   : num [1:75960] 0 0 0 0 0 0 0 0 0 ...
$ initialCrabCount   : int [1:75960] 0 1 0 0 0 0 1 0 0 ...
$ teamRiftHeraldKills : num [1:75960] 1 1 1 1 1 0 0 0 0 ...
$ earliestDragonTakedown : num [1:75960] 0 912 1612 912 912 ...
$ jungleKillEarlyJungle : num [1:75960] 0 0 0 0 0 0 0 0 0 ...
$ KillsonLanersEarlyJungleAsJungler : num [1:75960] 0 0 0 0 0 0 0 0 0 ...
$ KillsonLanersEarlyJungleAsLaner : num [1:75960] 0 0 0 0 0 0 0 0 0 ...
$ earlyLaningPhaseGoldExpAdvantage : num [1:75960] 1 0 0 0 0 0 0 0 0 ...
$ early_cs_per_min   : num [1:75960] 9.43 6.57 7.29 7.14 1.57 ...
$ early_gold_per_min : num [1:75960] 433 347 392 316 246 ...
$ jungle_proximity_pct : num [1:75960] 0.125 1 0.25 0.125 0.125 0.125 1 0.25 0.125 ...
$ roaming_pct        : num [1:75960] 0.2 0 0 0.1 0.1 0.3 0 0 0.2 0.2 ...
$ firstDragon        : num [1:75960] 0 0 0 0 0 0 1 1 0 ...
$ has_global_ult     : num [1:75960] 0 0 0 1 0 1 1 0 0 ...
```



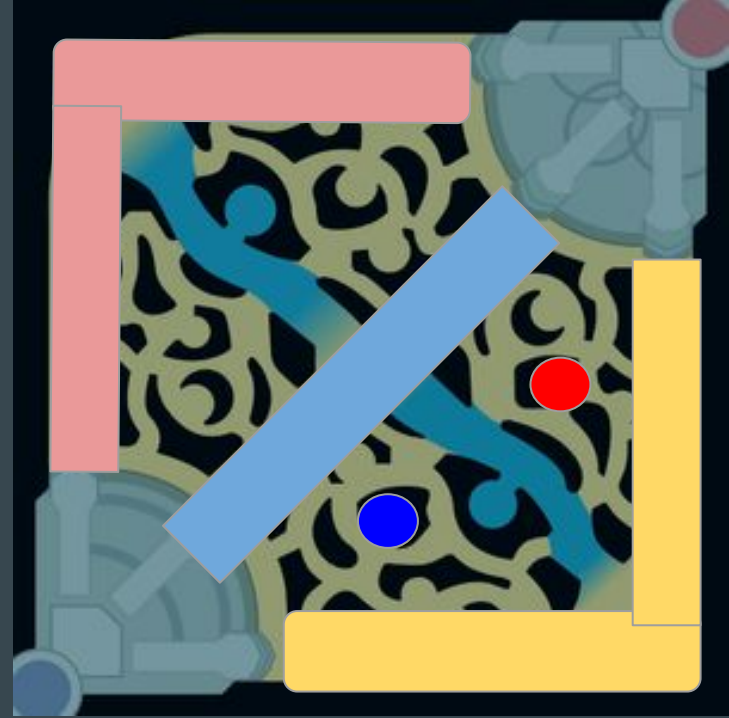


Considerations

- How much data to take
- Which metrics to include in the dataset
- “Exotic” picks

Derived Metrics

- Efficiency: gold/min - 18(cs/min)
 - Captures non-farming patterns
- Roaming Percentage
- Proximity Percentage
- Stomp Percentage

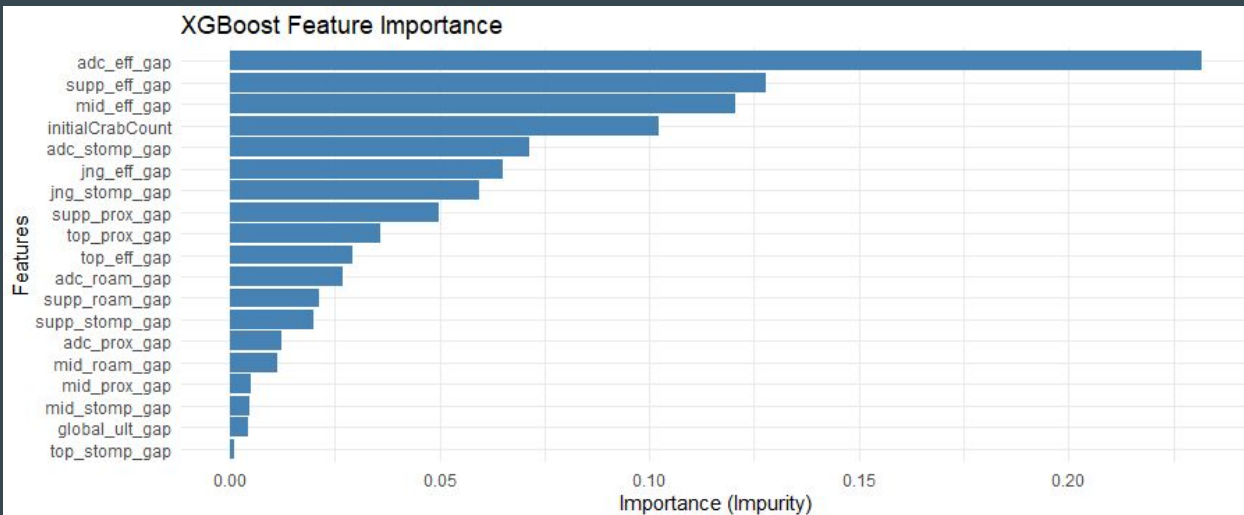


Prediction Model

- Data split into each team and then combined
- Model trained on leads in each metric for each role
 - Smoothed boosted tree for large number of predictions
- Raw stat leads considered, but dropped

```

match_level_training_set 995 obs. of 26 variables
$ matchId      : chr [1:995] "NA1_5564602956" "NA1_5566643292" "NA1_5568462341" "NA1_5568525062" ...
$ leader_won_drag : Factor w/ 2 levels "Leader_Drag",...: 2 1 2 1 2 2 2 1 2 1 ...
$ initialCrabCount : int [1:995] 1 1 1 1 1 1 1 1 0 ...
$ global_ult_gap   : num [1:995] 1 0 0 0 0 0 0 0 1 0 ...
$ top_gold_gap     : num [1:995] 89 154 126.86 -30 4.71 ...
$ jungle_gold_gap  : num [1:995] 125.1 -95.9 24.6 122 122.9 ...
$ mid_gold_gap     : num [1:995] -56.1 -28.9 -47.6 153.9 22.1 ...
$ adc_gold_gap     : num [1:995] 2.86 55.29 -33 41 119.43 ...
$ supp_gold_gap    : num [1:995] 17.43 -55.43 -3.71 34.57 56.14 ...
$ top_gold_eff_gap : num [1:995] 4.51 8.31 20.77 3.87 6.5 ...
$ jungle_gold_eff_gap : num [1:995] 12.11 -12.9 -2.53 6.73 10.56 ...
$ mid_gold_eff_gap : num [1:995] 36.13 4.44 -18.99 2.53 -1.03 ...
$ adc_gold_eff_gap : num [1:995] 1.618 -3.279 6.922 -0.897 -3.437 ...
$ supp_gold_eff_gap : num [1:995] 28.6 -159.7 -165.9 90 35.7 ...
$ top_stomp_gap    : num [1:995] 0 1 1 0 0 0 1 0 0 0 ...
$ jungle_stomp_gap : num [1:995] 0 0 0 1 1 0 0 0 0 0 ...
$ mid_stomp_gap    : num [1:995] 0 0 0 1 0 0 0 0 1 ...
$ adc_stomp_gap    : num [1:995] 0 0 0 0 1 0 0 0 0 ...
$ supp_stomp_gap   : num [1:995] 0 0 0 0 0 0 0 0 0 ...
$ top_prox_gap     : num [1:995] 0 0.125 0 0.125 0 -0.125 0 -0.125 0.25 -0.125 ...
$ mid_prox_gap     : num [1:995] -0.125 -0.25 -0.125 0 0.125 0 0.375 0 -0.125 0.125 ...
$ adc_prox_gap     : num [1:995] 0.25 0 -0.125 0 0 0 0.125 0 0 ...
$ supp_prox_gap    : num [1:995] 0.125 0 0 0.25 0 0.125 0.25 0.125 0 ...
$ mid_roam_gap     : num [1:995] 0 0 0 0 0 0 0 0.1 0 ...
$ adc_roam_gap     : num [1:995] -0.3 0.1 -0.3 0.3 -0.3 -0.2 -0.1 -0.1 0.3 0.1 ...
$ supp_roam_gap    : num [1:995] -0.3 0.3 -0.2 0.3 0 0 -0.1 -0.3 0.1 0.2 ...
    
```



Simulator

<https://jooba.shinyapps.io/finalapp/>

The screenshot displays the 'Simulator' interface, which is divided into two main columns for team selection and a bottom section for matchup metrics.

Blue Side Team Selection

- Top Lane: Select Champion
- Jungle: Select Champion
- Mid Lane: Select Champion
- ADC: Select Champion
- Support: Select Champion

Red Side Team Selection

- Top Lane: Select Champion
- Jungle: Select Champion
- Mid Lane: Select Champion
- ADC: Select Champion
- Support: Select Champion

Live Matchup Metrics

- First Dragon Control Probability**: A horizontal bar chart showing the probability for the Blue Side (blue) and Red Side (red).
- Predicted Team Efficiency Share**: A horizontal bar chart showing the efficiency share for the Blue Side (blue) and Red Side (red).
- Wackest Lane (Highest Stat Disparity): Even Matchup**

- Drafting page - create teams on each side
 - First dragon probability is target variable from the model
 - “Efficiency gap” captures the overall difference in early game pressure
 - “Wackest Lane” shows the probable target for early maneuvers
- Champion Stats page
 - Look at and sort through champions by role
- EDA page
 - Some visuals for reference during testing

Next Steps/Limitations

- Work with professional players/coaches create a stronger formula
- Wider scope
 - Comparisons between regions and ranks
- Target other objectives
- Live application to pull data while playing
- Data only taken from one region
- Riot Games API rate limits
- Possible omitted variables
- Difference between professional play and ranked solo queue